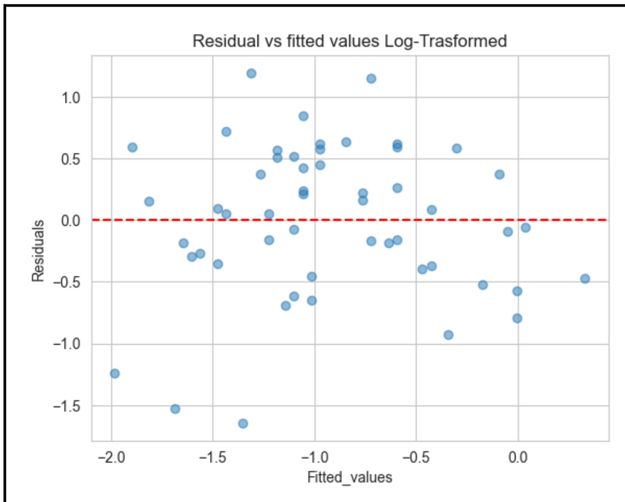
Tucker Inglefield
Math 3450
Final Case Study
04/23/2024

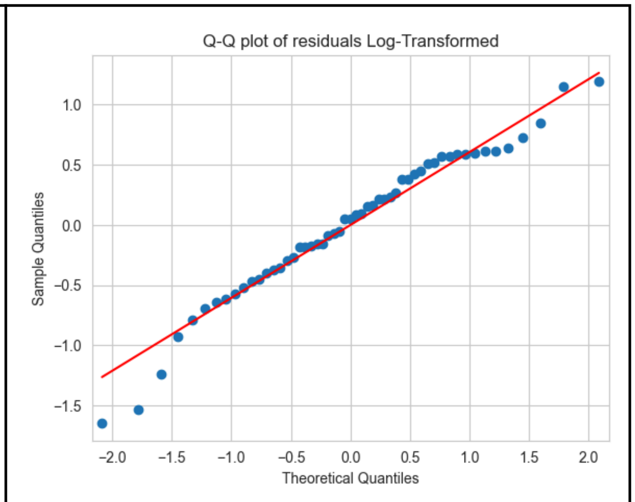
<u>Question 1:</u>

We were tasked with verifying or disputing 5 statements the original authors gave. We will address these statements one by one…

<u>*Statement 1*: Linear regression revealed that pH accounted for 41% of the variation in the standardized fish mercury concentration.</u>

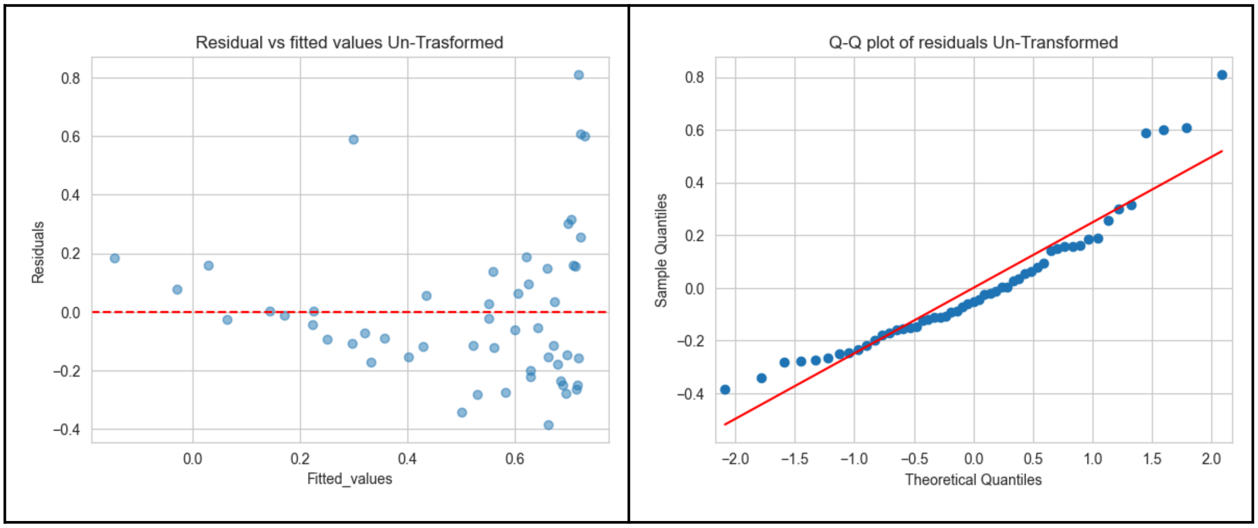
To verify this claim, we created a simple linear regression model to predict standardized fish mercury concentration from pH values. The r^2 value in this context would give us an interpretation like the statement. The untransformed data for StdMerc gave us an r^2 value of 0.363. After a log transformation on the response variable, we got an r^2 of 0.44. This was much closer to the original authors' claims. A residual vs fitted value graph and a qq-plot of this data showed that the assumptions of linear regression were well met. Because of this, we can agree with the authors' statement.





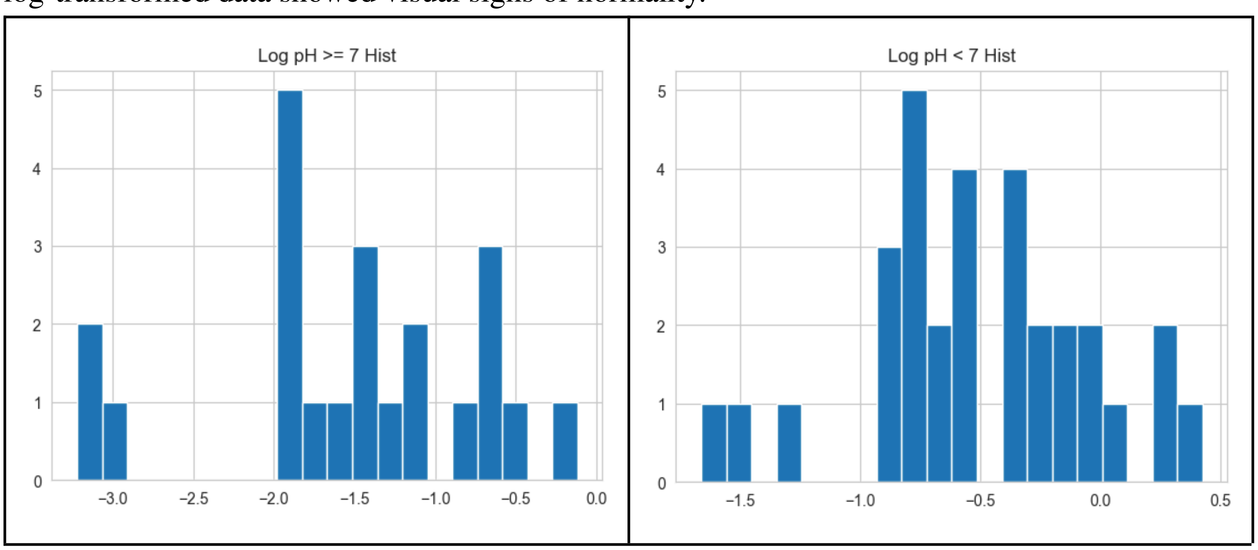
<u>*Statement 2*: Multiple regression revealed that chlorophyll *a* and alkalinity accounted for 45% of the variation in the standardized fish mercury concentration.</u>

To verify this claim, we created a model with Chlorophyll and Alkalinity as predictor variables. We suspected they might keep using the log transformed data, but when we ran the model with logged values for our response variable, we got an adjusted r^2 of 0.735. When we ran the model with the untransformed data, we found their claim with an r^2 of 0.45. The same plots used in statement 1, showed that the assumptions for multiple regression were not well met, with unequal variances and questionable normality. We can understand where they got their claim, but do not necessarily agree with it, because of the violation of assumptions.



<u>*Statement 3*: Mercury concentrations were significantly higher in lakes w/ pH less than 7.</u>

We first created groups of data based on the authors' claims. One of lakes with pH less than 7, and another with pH >= 7. We wished to conduct a two sample t-test to verify the authors claims. The untransformed data fulfilled the assumption of equal variances, with a Levene test result giving a p-value of 0.086. The untransformed data did not satisfy the assumption of normality with p-values of 0.01 and 0.02. We could use the large sample size to buy normality but we proceeded with a log transformation since the original authors appeared to use one. This log transformation on the response variable allowed us to verify the assumption of equal variance(through a Levene's p-value of 0.061) and the assumption of normality(with shapiro p-values of 0.08 and 0.65 for the groups). Furthermore, histograms generated with the log-transformed data showed visual signs of normality.



With our assumptions met, we proceeded with a two sample t-test with equal variance. This gave us a p-value of 8.63*10^-0.7. This gave us significant evidence that the median standard mercury concentration was different among the groups we created. A simple calculation of averages among these two groups' stand concentration verified the directionality of the author's claim as well.

<u>*Statement 4*: Mercury concentrations were significantly higher in lakes w/ Alkalinity less than 20 mg/L.</u>

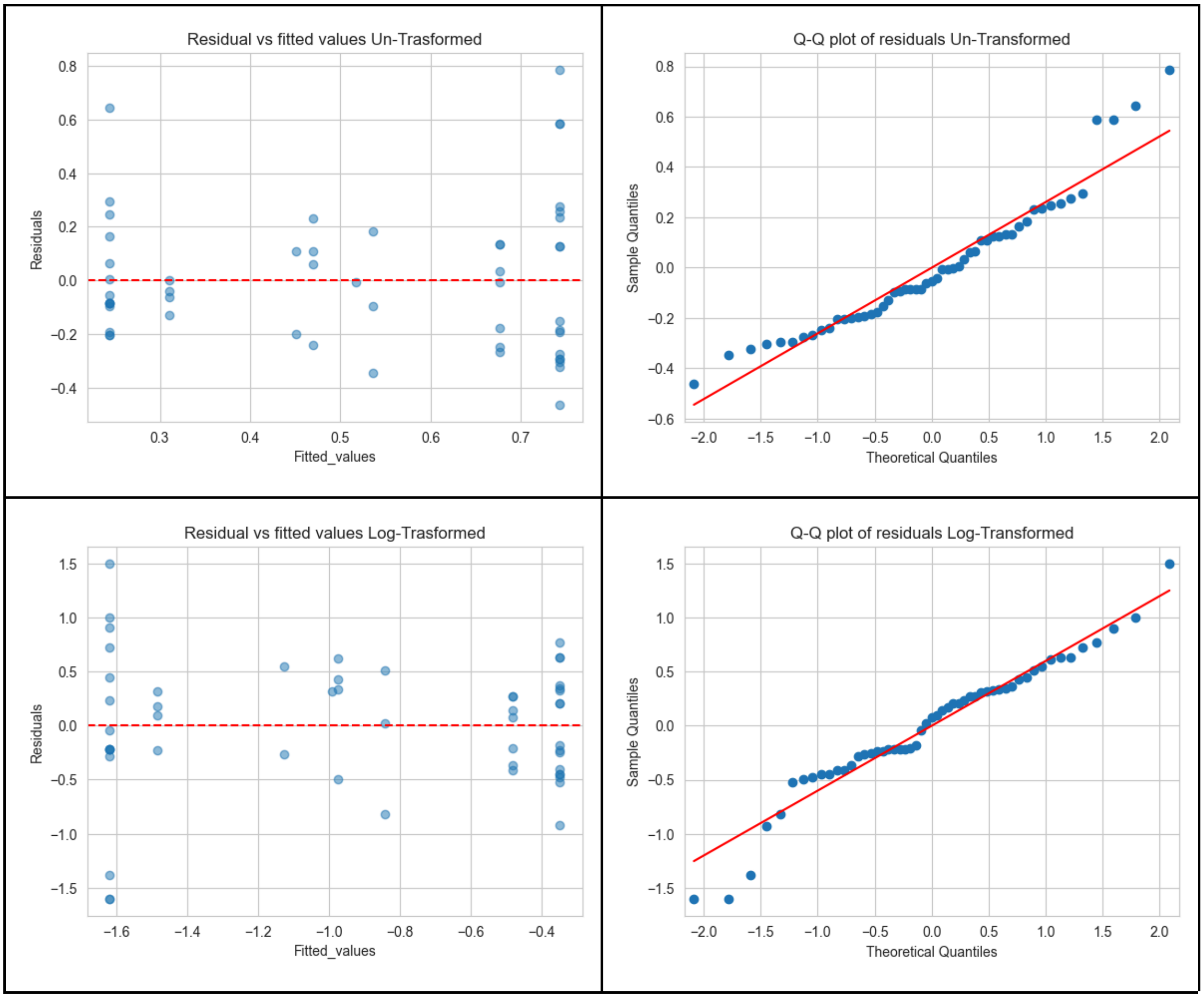
We repeated our process from statement 3; creating the relevant groups and testing the assumptions on the untransformed and log-transformed data, with the same goal of a two sample t-test. The untransformed data violated the assumption of normality (with Shapiro obtained p-values of 0.02 and 0.01) but passed the assumption of equal variances (with a Levene obtained p-value of 0.176). The log-transformed data passed the assumption of normality (with Shapiro obtained p-values of 0.05 and 0.58) but violated the assumption of equal variances (with a Levene obtained p-value of 0.02). Due to the sample size being less than 30, we are unable to use this to buy normality. Knowing this, we proceeded with an Welch T-test of unequal variances, choosing to favor the normality of the log-transformed data. This test gave us a p-value of 1.0*10^-5. We again repeated our average calculation and confirmed directionality as well.

<u>*Statement 5*: Mercury concentrations were significantly higher in lakes w/ Chlorophyll less than 10 mug/L.</u>

We again repeated our same process from statements 3 and 4. We were unable to get both groups to be normally distributed at the same time, and as previously stated we could not use the sample size to buy normality. Because of this, we proceed with a non-parametric test, specifically the Mann-Whitney-U test. This test gave us a p-value of 0.015, allowing us to verify the authors' claim. Again, calculation of averages among groups verified the directionality.

<u>Question 2:</u>

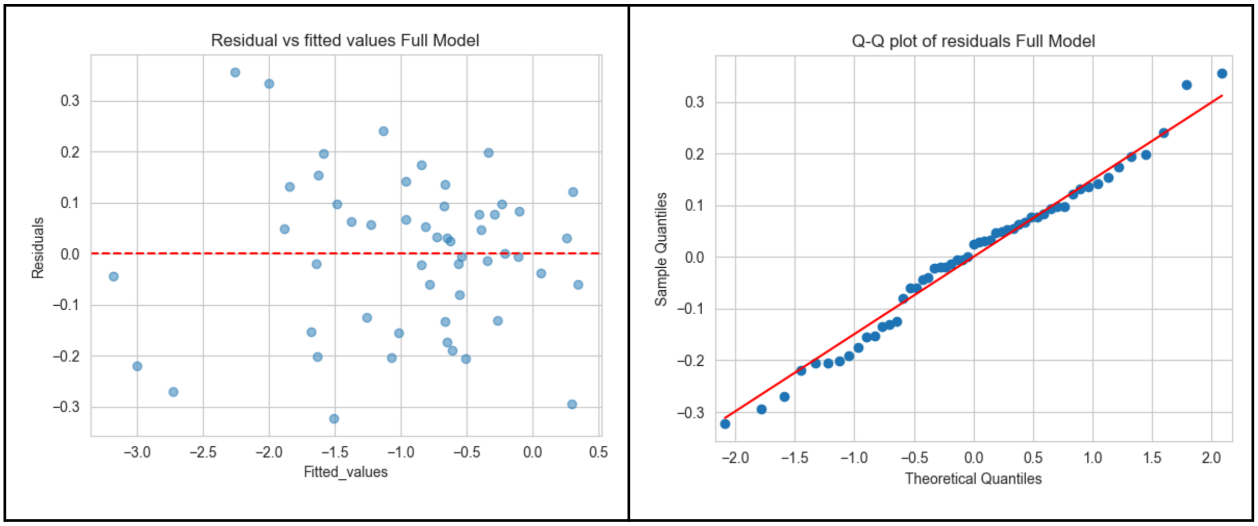
To assess the strength of these categorical attributes, we created dummy variables and ran a multiple regression model. We created dummy variables for the cases outlined in the previous statements: low_chlor where chlorophyll was less than 10, low_alk where alkalinity was less than 20 and low_pH where pH was recorded less than 7. We fit the model using an untransformed and log-transformed response variable.



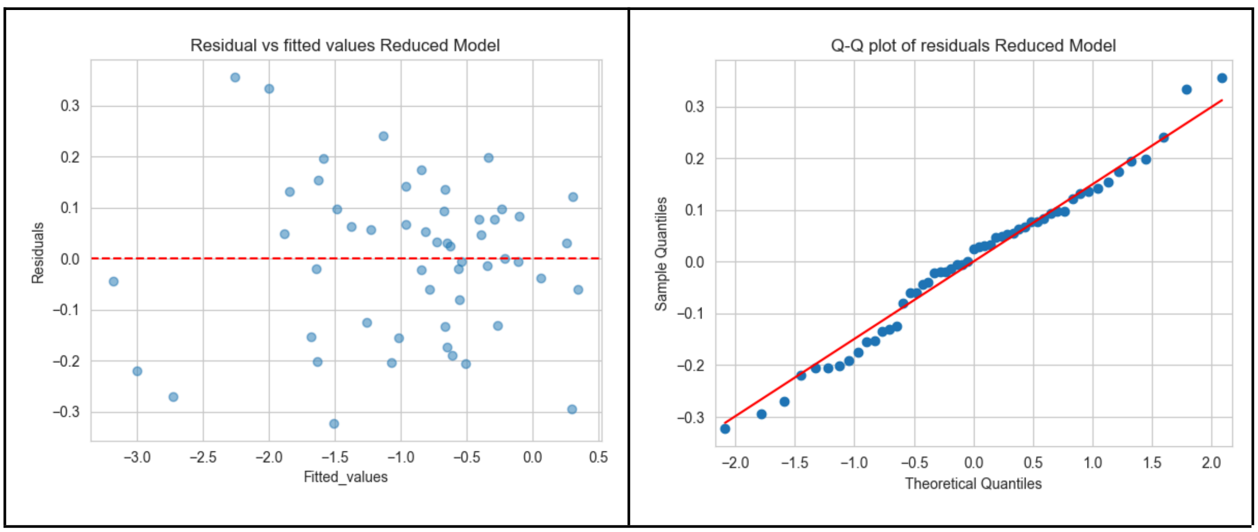
The log transformation improved the assumption of normality visually with the QQ plot and the assumption of equal variance appeared not to be affected by the transformation, at least not significantly visually. The untransformed and log-transformed models gave us adjusted r^2 values of 0.357 and 0.416 respectively. For the sake of assumptions and the higher explainable portion of variability among the response variable, we chose the log transformed model. In the best case however, these categorical variables were not able to explain a large proportion of the variability in the response variable. They will be considered for future model building, but not much else.

Question 3:

For model development, we created a full model with all predictors including the dummy variables from question 2. This model gave us an adjusted R^2 of 0.947 and the following residual and QQ-plots:



We used backwards selection to get to a reduced model by eliminating the least significant predictor sequentially, until all predictors were below the standard alpha of 0.05. Our final model had these predictors: Alkalinity, pH, Chlorophyll, numSamples, min & max. This model gave us an adjusted R^2 of 0.947 with the following plots:



We also conducted an ANOVA to test for a significant difference between the two models. As we can see from the output, there is likely not a significant difference between the two models. This can further be corroborated by the identical R^2 and plots. This means that the terms we gained in the full model were not significantly impactful on the model as a whole, and the reduced model had much better significance in its predictors, per the regression output. We also considered interaction terms, but due to the large number of predictors, including every interaction term lead to overfitting the model. We did not see a dramatic difference in R^2 when including individual interaction terms as well.

```
red_model = ols("LogStdMerc ~ Alkalinity + pH + Chlorophyll + numSamples + min + max ", data = df)
full_results = red_model.fit()
print(full_results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              LogStdMerc   R-squared:                       0.953
Model:                             OLS   Adj. R-squared:                  0.947
Method:                  Least Squares   F-statistic:                     155.4
Date:                 Mon, 22 Apr 2024   Prob (F-statistic):           7.94e-29
Time:                         18:38:25   Log-Likelihood:                 17.044
No. Observations:                   53   AIC:                            -20.09
Df Residuals:                       46   BIC:                            -6.296
Df Model:                            6
Covariance Type:             nonrobust
===============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------
Intercept     -1.8672      0.218     -8.583      0.000      -2.305      -1.429
Alkalinity    -0.0061      0.001     -5.708      0.000      -0.008      -0.004
pH             0.0903      0.033      2.711      0.009       0.023       0.157
Chlorophyll   -0.0114      0.001    -10.396      0.000      -0.014      -0.009
numSamples     0.0090      0.003      2.729      0.009       0.002       0.016
min            1.2687      0.194      6.529      0.000       0.878       1.660
max            0.4165      0.092      4.537      0.000       0.232       0.601
==============================================================================
Omnibus:                         1.550   Durbin-Watson:                   1.726
Prob(Omnibus):                   0.461   Jarque-Bera (JB):                1.380
Skew:                           -0.386   Prob(JB):                        0.501
...
==============================================================================
```

```
full_model = ols("LogStdMerc ~ Alkalinity + pH + Calcium + Chlorophyll + Avg_Mercu
full_results = full_model.fit()
red_model = ols("LogStdMerc ~ Alkalinity + pH + Chlorophyll + numSamples + min + m
red_results = red_model.fit()
```

```
# Use anova to compare full vs reduced model...
from statsmodels.stats.anova import anova_lm
anova_results = anova_lm(red_results, full_results)
print(anova_results)
```

```
   df_resid       ssr  df_diff   ss_diff         F    Pr(>F)
0      46.0  1.631072      0.0       NaN       NaN       NaN
1      40.0  1.419848      6.0  0.211224  0.991768  0.443968
```

Question 4:

Again, we will compare/contrast our findings with the original authors' statements individually. These questions asked for correlations of individual explanatory variables, which we did not initially consider in our models. For the sake of answering this question we have calculated these. Due to the extent of our current programming knowledge, we were able to calculate these values as follows, but lost the directionality, which we will find in the coefficients of our linear regression model.

```
# Correlation Comparison
explanatory_vars =['Alkalinity', 'pH', 'Calcium', 'Chlorophyll',
'numSamples', 'min', 'max', 'age_data', 'low_chlor', 'low_alk', 'low_pH']
for i in explanatory_vars:
    i_model = ols(f"LogStdMerc ~ {i}", data = df)
    i_results = i_model.fit()
    i_r_sq = i_results.rsquared_adj
    i_corr = np.sqrt(i_r_sq)
    print(f'{i} Corr: {i_corr}')
```

```
Alkalinity Corr: 0.7255832299487426
pH Corr: 0.6549118571749947
Calcium Corr: 0.5484101683741908
Chlorophyll Corr: 0.7456270405969193
numSamples Corr: nan
min Corr: 0.7876256500281476
max Corr: 0.8495081292264084
age_data Corr: 0.21831528323477956
low_chlor Corr: 0.3411193692001917
low_alk Corr: 0.5721391497919728
low_pH Corr: 0.6071188798545563
/var/folders/yl/0k738k9x1wn7tbbplx4xpf6c0000gn/T/ipykernel_46541/2249452708.py:8: R
  i_corr = np.sqrt(i_r_sq)
```

- *"Total mercury concentrations in axial muscle of individual fish ranged from 0.04 to 2.04 µg/g wet weight and were positively correlated with fish age (strongest correlation) and size."*

We did not find age to be the highest correlation of all predictor variables. In fact, it was not even in our reduced model, so in our experiments, this variable was not significant.

- *"There was no difference in the rate of mercury accumulation by age between the sexes, even though females grew faster."*

We were not given access to a sex variable in the dataset provided so we have no way of comparing this claim to our results.

- *"Chemical characteristics of lakes strongly influenced the bio-accumulation of mercury in large mouth bass."*

This is a tricky one. We would be more careful about the wording. While we can look at our model and say that the results were *statistically* significant (high R^2, good validation of assumptions, significant predictor p-values, etc), we lack the domain knowledge to claim that the chemical characteristics are significant in fish. Amounts are tricky. Something may seem large proportionally but when put in context of the domain is actually not. Take for example an earlier case study, where we found that men made on average 1$ more than female professors. This was on a *salary* variable and not an hourly one, so the significance of this outcome was purely statistical.

- *"Mercury concentrations, standardized to age-3 fish for comparison among lakes, ranged from 0.04 to 1.53 µg/g and were negatively correlated with alkalinity, calcium, chlorophyll a, conductance, magnesium, pH, total hardness, total nitrogen, and total phosphorus."*

Some of these variables were not included in the dataset so we can only speak to the ones we had access to. Alkalinity, Calcium and Chlorophyll all had negative coefficients in our full model so we can agree with this part of the interpretation. We found that pH had a positive coefficient and so would most likely have a positive correlation, so we cannot agree with this statement.

- *"No significant correlations were observed between the standardized fish mercury concentration and lake color, secchi transparency, tannic acid, or surface area."*

Again, these variables are not given in the dataset provided so we have no comment on these claims.